Batteries

Seaglider's main battery pack is not just an energy store — it **is the trim mechanism**. The large main battery, with a brass weight bolted to its underside, is the mass that the mass shifter slides fore/aft for pitch and rotates for roll. That double duty means every battery change touches almost everything else: vehicle mass, ballast, pitch/roll trim, and even the compass calibration.

Source

Paraphrased from the APL-UW IOP *SGX Documentation* (v1.0, 2024), the Electrochem primary-lithium *Safety and Handling Guidelines*, and the battery MSDS/SDS sheets (UN3090/UN3091). Pack configurations vary by vehicle generation and build — defer to APL-UW IOP and your battery vendor's documentation.

Pack architectures by generation

Generation	Packs	Notes
Legacy SG (1KA era)	One **24 V** + one **10 V** lithium primary	24 V pack drives the motors (VBD, mass shifter) and is the moving trim mass; 10 V pack lives under the main electronics board and powers electronics
SG (univolt)	15 V packs, 12.0 kg total, 0.56 kg lithium	Effective capacity ~350 Ah; the 10 V/24 V distinction disappears (but voltage-select jumpers must still be installed)
SGX	Three 15 V packs, 19.6 kg total, 0.91 kg lithium	Effective capacity ~575 Ah; larger forward pack plus an additional aft pack behind the mass shifter

On all generations the main pack carries an **1100 g brass weight** on its bottom face — the axial asymmetry that makes roll control work when the pack is rotated (±80° on SGX, ±40° on SG).